PAPER_937: Blazar Multi-Messenger Phonon Correlation

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 212 **Source:** blazar_jet_phonon.py (BlazarMultiMessengerPhononCorrelation) **Calculator:** BlazarMultiMessengerPhononCorrelationCalc (CP4 #521) **CVW:** v2.0.0 compliant

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Abstract

We derive the multi-messenger (VHE gamma-ray and neutrino) luminosity correlations for blazars with UQFF phonon-enhanced pair cascades. The VHE gamma-ray luminosity scales as L_VHE proportional to P_jet (1 + Phi S_26) delta_D^4, while the neutrino luminosity is L_nu proportional to L_VHE f_pg (1 + Phi S_26 [SSq]/N). The phonon enhancement factor (1 + Phi S_26) boosts both channels relative to standard BZ predictions, with the relative neutrino/gamma-ray ratio providing a diagnostic for the phonon coupling strength.

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1. Core Equations

VHE gamma-ray luminosity (phonon-enhanced):

$$L_{\mathrm{VHE}} = P_{\mathrm{jet}} \cdot (1 + \Phi \cdot S_{26}) \cdot \delta_D^4$$

Neutrino luminosity (phonon-enhanced):

$$L_\nu = L_{\mathrm{VHE}} \cdot f_{p\gamma} \cdot \left(1 + \frac{\Phi \cdot S_{26} \cdot [\mathrm{SSq}]}{N}\right)$$

where: - $P_{\mathrm{jet}} = P_{\mathrm{BZ}}(1+M_{\mathrm{jet}})$ is the phonon-modulated jet power - $f_{p\gamma} \sim 0.05$ is the photopion production efficiency - $N = 26$ is the number of UQFF layers - δ_D is the Doppler factor from bulk Lorentz factor Gamma_bulk

Enhancement relative to standard BZ:

$$\frac{L_{\mathrm{VHE}}^{\mathrm{UQFF}}}{L_{\mathrm{VHE}}^{\mathrm{BZ}}} = \frac{(1 + M_{\mathrm{jet}})(1 + \Phi S_{26})}{2}$$

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2. UQFF Integration

The `BlazarMultiMessengerPhononCorrelationCalc` (CP4 #521) computes P_jet, L_VHE, L_nu, and delta_D for arbitrary blazar parameters. The simulate() method sweeps Gamma_bulk = [5, 10, 15, 20, 30] to map the Doppler-dependent multi-messenger signal.

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3. Physical Significance

The IceCube detection of high-energy neutrinos coincident with the blazar TXS 0506+056 opened the era of multi-messenger blazar astronomy. The UQFF phonon enhancement provides a mechanism to explain the observed neutrino-to-gamma-ray ratio: the (1 + Phi S_26) factor boosts both channels but with different scaling, creating a characteristic spectral signature testable with future IceCube-Gen2 and CTA observations.

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4. Source Data

- **File:** blazar_jet_phonon.py
- **Session:** 212
- **CP4 Class:** BlazarMultiMessengerPhononCorrelationCalc (#521)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. IceCube Collaboration – Neutrino emission from the direction of the blazar TXS 0506+056, Science 361, 147 (2018)
3. IceCube Collaboration – Multimessenger observations of a flaring blazar coincident with high-energy neutrino IceCube-170922A, Science 361, eaat1378 (2018)